

3	A satellite of mass 2400 kg is projected from the Earth's surface and placed in a geostationary orbit at a distance of 4.23×10^7 m from the centre of the Earth. The Earth may be assumed to be a uniform sphere and has mass of 6.00×10^{24} kg and radius of 6.37×10^6 m.	
	(a) State two conditions for the satellite at this distance to be considered a geostationary satellite. 	[2]
	(a) Solution: The orbit of the satellite lies in the equatorial plane. The satellite orbits from West to East. The period of rotation is 24 hours.	1 1
	(b) Explain why the satellite does not move in the direction of the gravitational force. 	[2]
	(b) Solution: The gravitational force provides the centripetal force, and the force is perpendicular to the velocity of satellite,	1 1

		which causes the satellite to experience a change in the direction of motion and keep its distance from the Earth constant. It therefore moves in a circular motion and does not change its speed.	
	(c)	Calculate the change in gravitational potential energy of the 2400 kg satellite when it is placed in the geostationary orbit from the Earth's surface. change in gravitational potential energy = J	[3]
	(c)	Solution: $\Delta G.P.E. = -\frac{GMm}{r_f} - \left(-\frac{GMm}{r_i}\right) = -GMm \left[\frac{1}{r_f} - \frac{1}{r_i}\right]$ $\Delta G.P.E. = -\left(6.67 \times 10^{-11}\right) \times \left(6.00 \times 10^{24}\right) \times (2400) \times \left(\frac{1}{4.23 \times 10^7} - \frac{1}{6.37 \times 10^6}\right)$ $\Delta G.P.E. = 1.28 \times 10^{11} \text{ J}$	1 1 1
	(d)	Due to a technical fault in the satellite, it is regarded as a space junk in the geostationary orbit. With reference to your answer in (c), suggest a reason why it is considered a possible safety hazard for the satellite to fall back to Earth. 	[2]
	(d)	Solution: There will be a large change in the satellite's gravitational potential energy as it falls back to Earth. The change in potential energy will be converted into a gain in the satellite's kinetic energy. The satellite will acquire a high speed. Due to friction with air particles at high speed, the satellite may heat up and hence burn up into many fragments upon entering the Earth's atmosphere.	1 1

4	(a)	A beam of light from a sodium lamp passes through a pair of narrow slits S_1 and	
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